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Kamlet–Taft Parameters of Deep Eutectic Solvents and Their Relationship with Dissolution of Main Lignocellulosic Components

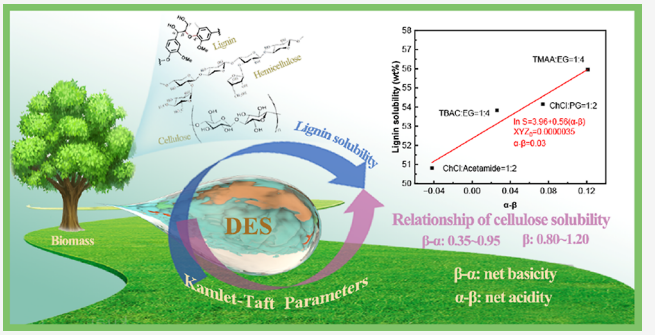
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Cite This: *Ind. Eng. Chem. Res.* 2023, 62, 11723–11734

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ABSTRACT: Deep eutectic solvents (DESs) have attracted significant interest in dissolving lignocellulosic components. However, revealing the relationship between the solvation properties of DESs and their ability to dissolve lignocellulosic components remains a significant challenge. Herein, 56 DESs were prepared and applied to dissolve lignocellulosic components. This revealed that the Kamlet–Taft parameters (i.e., α, β, and π^*) were significantly affected by the structures of DESs. More notably, the correlation between Kamlet–Taft parameters and solubility varied with DESs and lignocellulosic components. Concretely, DESs have greater values of β and $\beta-\alpha$ (net basicity), are expected to dissolve cellulose efficiently, have greater net basicity, and are favorable to dissolve lignin, and the solubility of xylan was linearly correlated with the β value. These findings provided valuable information and necessary theoretical guidance on designing robust DESs to dissolve lignocellulosic components based on Kamlet–Taft solvatochromic parameters.



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1. INTRODUCTION

Lignocellulose, composed of three primary lignocellulosic components of cellulose, hemicellulose, and lignin, is the largest renewable resource on earth. It can be used as an abundantly available raw material to produce value-added products, including transportation fuels, platform chemicals, and functional materials.[1,2] Generally, dissolution is indispensable to the efficient utilization or conversion of lignocellulose. Significant effort has been devoted to developing suitable solvents to dissolve lignocellulosic components, a research hotspot in the field of lignocellulose.[3] Of the developed solvent systems, deep eutectic solvents (DESs) have been recognized as an emerging class of functional solvents for the dissolution of lignocellulosic components. Generally, DESs are composed of two components via the hydrogen bonding interaction to generate a eutectic mixture, and the two components are hydrogen-bond donor (HBD) and hydrogen-bond acceptor (HBA), respectively.[4,5] The DESs designed by selecting suitable HBDs and HBAs of different properties could enable them to possess various unique advantages, e.g., strong solvation ability, good designability, low cost, facile preparation, high biocompatibility, etc.[4−8] These characteristics of DESs promote the continually growing interest in constructing efficient DESs to dissolve cellulose, hemicellulose, or lignin.[9−12] As reported, DESs can be defined as the mixture of two or more pure compounds with a eutectic point temperature below that for an ideal liquid mixture, showing significantly negative deviations from ideality.[13] More importantly, the composition of DESs should be in a certain range, not just in a fixed stoichiometric proportion.[13] Based on this definition, all the prepared solvents in this work can be considered as DESs. Regarding the utilization of DESs in lignocellulosic chemistry, the current attention is mainly focused on designing novel DESs for the pretreatment of lignocellulose or dissolution of the three major components.[14] Very little attention has been paid to systematically analyze the relationship between the properties of DESs and the dissolution of lignocellulosic components, resulting in limited rules that can be followed to design or screen suitable DESs to dissolve cellulose, hemicellulose, and lignin.[15]

As is well known, whether a DES is a good solvent or not is mostly related to its solvation ability. Generally, the solvation properties of one specific solvent can be evaluated by Kamlet–Taft solvatochromic parameters (i.e., α, β, and π^*), in which α represents the hydrogen-bond-donating ability (i.e., Lewis acidity), β represents the hydrogen-bond-accepting ability (i.e., Lewis basicity), and π^* represents the polarizability of the solvents.[16] Some attempts have been employed using Kamlet–Taft parameters to evaluate the performance of DESs on

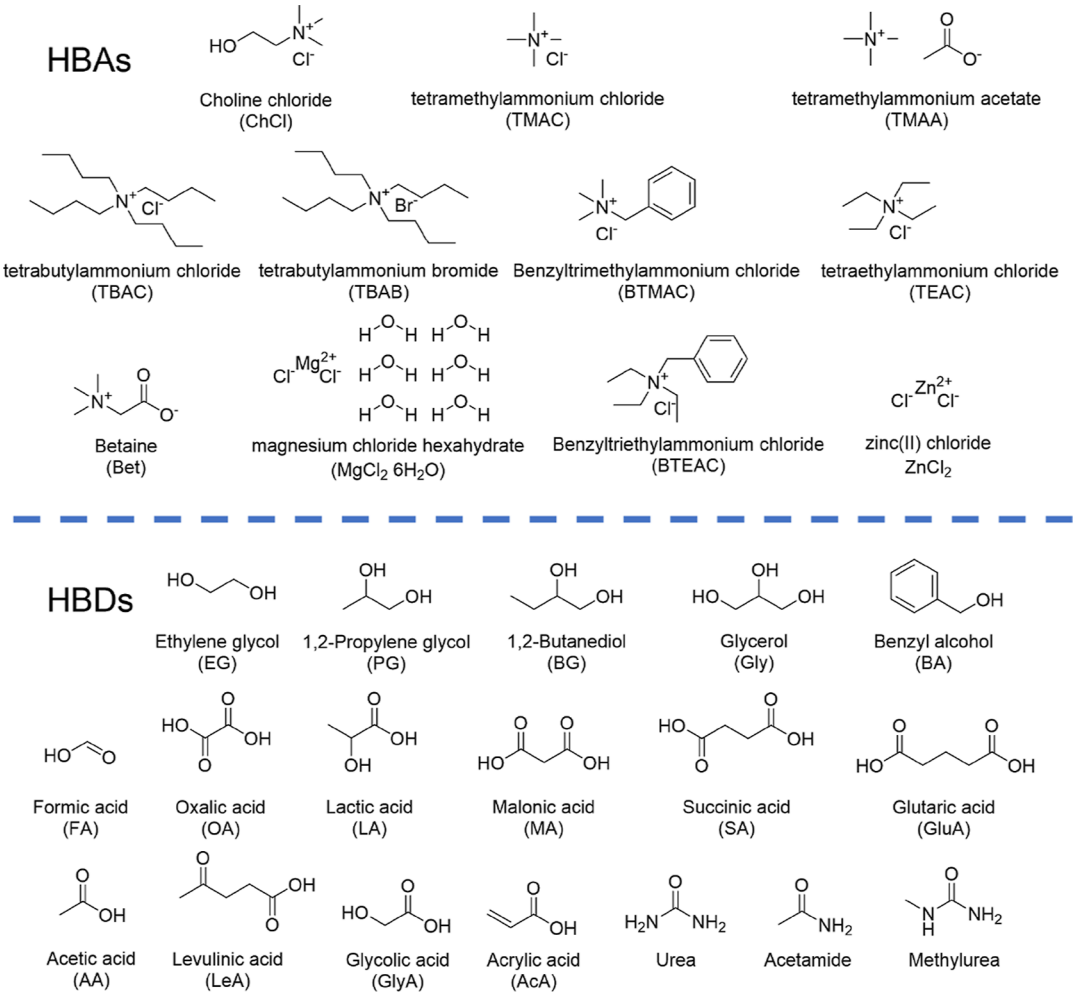
Received: April 20, 2023
Revised: June 29, 2023
Accepted: July 6, 2023
Published: July 18, 2023



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https://doi.org/10.1021/acs.iecr.3c01309
Ind. Eng. Chem. Res. 2023, 62, 11723−11734

Scheme 1. Chemical Structures of the Employed HBDs and HBAs



pretreatment of lignocellulose and dissolution of lignocellulosic components. Hong et al. found that a lower lignin yield was achieved from the pretreatment of poplar in choline chloride/oxalic acid (ChCl/OA) than ChCl/lactic acid (ChCl/LA),[17] probably resulting from the higher $\alpha-\beta$ value of ChCl/OA (1.10) than that of ChCl/LA (0.80). Liang et al. utilized 10 types of LA-based DESs to pretreat corn straw[18] and observed a positive correlation between α and the pretreatment performance. Moreover, a positive linear relationship between α and the extraction efficiency of lignin was found when using different ChCl-based DESs to pretreat pine wood.[19] In addition, our previous works showed that the solubility of lignin had a positive correlation with α and π^* of the DESs, while the solubility of xylan had a positive correlation with β.[11,20]

However, previous works only investigated the relationship between Kamlet−Taft parameters of limited types of DESs and their performance on the pretreatment/dissolution of lignocellulose. More difficultly, although Kamlet−Taft parameters of some DESs have been experimentally determined or theoretically predicted,[21] the reported parameters of DES are still highly limited compared with the great amount of DESs, let alone to numerous possible combinations of DESs which have not been investigated.[22] Besides, the parameters reported in different works were not obtained by compatible methods, leading to the incomparability of these parameters. Owing to the insufficient and even incompatible data, Kamlet−Taft solvatochromic parameters were still not the good indicators on the solute solubility in comparison with the surface tension and pK_a. Therefore, it is a great challenge to evaluate the performance of DES on the dissolution of lignocellulosic components using the Kamlet−Taft parameters.[23] More effort should be devoted on investigating the correlation between the structures of DESs and Kamlet−Taft parameters and the relationship between Kamlet−Taft parameters and the solubility of lignocellulosic components.

To better understand the Kamlet−Taft parameters of DESs to facilitate the design of DESs to dissolve lignocellulosic components, 56 kinds of DESs were prepared using different HBAs and HBDs. Emphasizing, these DESs were selected based on the properties of HBAs and HBDs. Some previously reported DESs were directly selected, and most of the involved DESs were prepared based on our continuous jobs on the preparation of novel DESs to dissolve the lignocellulosic components. Initially, the influence of the structure (chain length) and properties of HBDs and HBAs on Kamlet−Taft parameters was analyzed. Additionally, the Kamlet−Taft parameters of DESs were compared with those of conventional solvents and ionic liquids (ILs), providing a macroscopic image of Kamlet−Taft parameters of different solvents. Furthermore, the relationship between the Kamlet−Taft parameters of the DESs and the solubility of lignocellulosic

https://doi.org/10.1021/acs.iecr.3c01309
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components was investigated for better understanding the influence of DESs on the dissolution of lignocellulosic components. These results would help design robust DESs or simplify the selection process of DESs for dissolution of lignocellulosic components. Herein, we should point out that not all the used DESs were non-toxic by nature. The toxicity of DESs depends on the components used to form DESs. However, with the development of DESs, non-toxic and safe DESs, possessing the most suitable Kamlet−Taft parameters, could be constructed for dissolving lignocellulosic components.

2. EXPERIMENTAL SECTION

Chemicals used for the preparation of DESs were obtained from Beijing Innochem Technology Co., Ltd (Beijing, China), Energy Chemical (Beijing, China), J&K Scientific, Fluorochem (Old Glossop, U.K.), Acros Organic, and Sigma-Aldrich (Shanghai, China), separately. The Kamlet−Taft solvatochromic parameters for DESs and the solubility of lignin/xylan/cellulose in DESs were determined according to our previous work.[10] The Kamlet−Taft linear solvation energy relationship (LSER) method was used to determine the correlation between solubility and Kamlet−Taft parameters.[24,25] The detailed procedures are provided in the Supporting Information section.

3. RESULTS AND DISCUSSION

3.1. Measurement of Kamlet−Taft Solvatochromic Parameters of DESs. The Kamlet−Taft solvatochromic parameters of the involved DESs (56 kinds, Scheme 1) were determined by the maximum UV−vis absorption peaks of the solutions of three dyes (Figure S1). The corresponding Kamlet−Taft parameters was the average of twice the measurements for each DES obtained from twice preparations (Table S2). Herein, it was emphasized that the Kamlet−Taft parameters of different solvents were determined using the same dyes (i.e., *N*,*N*-diethyl-4-nitroaniline, 4-nitroaniline, and Nile red) to exclude the influence of different dyes on the values of Kamlet−Taft parameters and make the results be comparable. By analyzing the values of α, β, and π^*, we could conclude the relationship between the values of Kamlet−Taft solvatochromic parameters and the structures and properties of DESs. Furthermore, by combining the solubility of cellulose, lignin, and hemicellulose in DESs, the correlation between solubility and Kamlet−Taft parameters could be revealed to some extent.

3.2. Influence of DESs' Structures on the Kamlet−Taft Solvatochromic Parameters. *3.2.1. Effect of HBAs.* Initially, we studied the effect of HBAs on Kamlet−Taft parameters. Herein, the DESs were composed by ethylene glycol (EG) as the HBD and eight compounds as the HBAs with a molar ratio of HBD and HBA of 1:4. As shown in Figure 1, $ZnCl_2$/EG—1:4 has the largest π^* (polarity) and α (hydrogen bond acidity) values but the smallest β (hydrogen bond basicity) value among the DESs because $ZnCl_2$/EG—1:4 is probably formed through an oxygen bond between the O atom in EG and the metal center of $ZnCl_2$ rather than hydrogen bonds.[26] In comparison, the difference in the values of α, β, and π^* for other seven DESs was not significant because the HBAs in these DESs were all quaternary ammonium salts. Additionally, TMAA/EG—1:4 showed the lowest values of α, β, and π^* among the DESs formed between EG and quaternary ammonium salts, probably resulting from the different anions (acetate vs halogen ions).

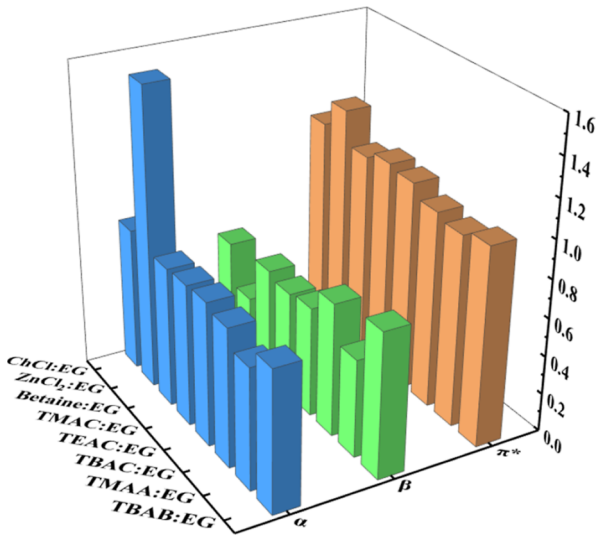


Figure 1. Effect of the types of HBAs on the values of α, β, and π^*.

It has been reported that $ZnCl_2$-based DESs could be generated between $ZnCl_2$ and some amides or alcohols, while ambient temperature DESs could not be formed between $ZnCl_2$ and carboxylic acids.[26] However, in this work, we found that one DES could be formed between $ZnCl_2$ and LA (LA, a kind of carboxylic acid) with a molar ratio of 1:2 owing to the existence of a hydroxyl group in the structure of LA, which was also consistent with the fact that DESs could be formed between $ZnCl_2$ and alcohols.[26] To further analyze the effect of quaternary ammonium salts and metal halides on Kamlet−Taft parameters, three groups of DESs were constructed using ChCl and $ZnCl_2$ as the HBAs and EG, urea, and LA as the HBDs, separately. The molar ratio of HBA and HBD was the same for the DESs with EG and LA as the HBDs, while it was 1:4 and 1:2 for $ZnCl_2$/urea and ChCl/urea, respectively. As shown in Figure 2, metal halide-based DESs (i.e., $ZnCl_2$/EG—1:4 and $ZnCl_2$/urea—1:4) showed a higher value of π^* than ammonium salt-based DESs (i.e., ChCl/EG—1:4 and ChCl/

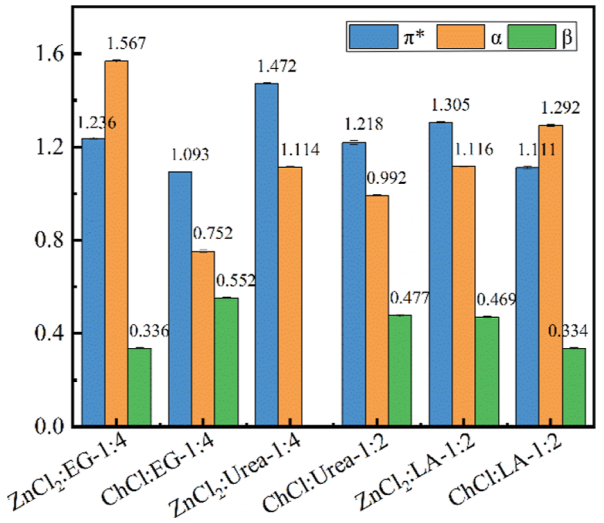


Figure 2. Comparison of α, β, and π^* values of $ZnCl_2$- and ChCl-Based DESs.

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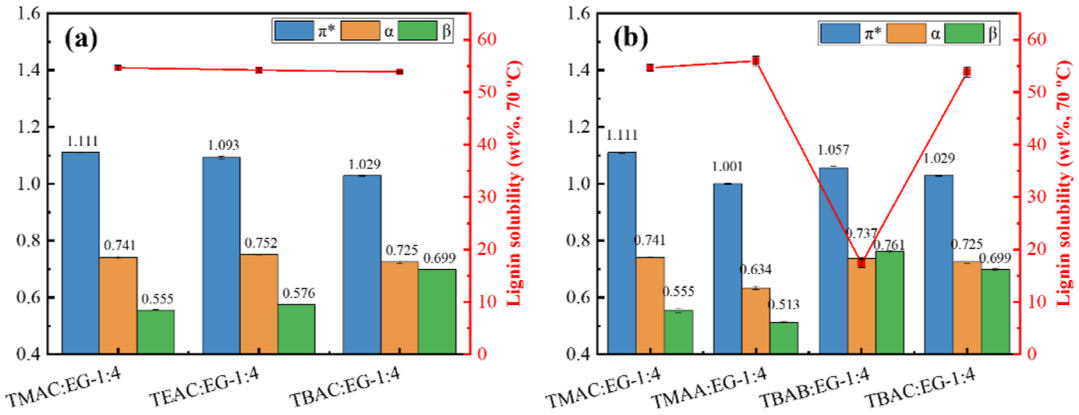


Figure 3. Effect of the alkyl chain length (a) and the anion type (b) of HBAs on the values of α, β, and π^*.

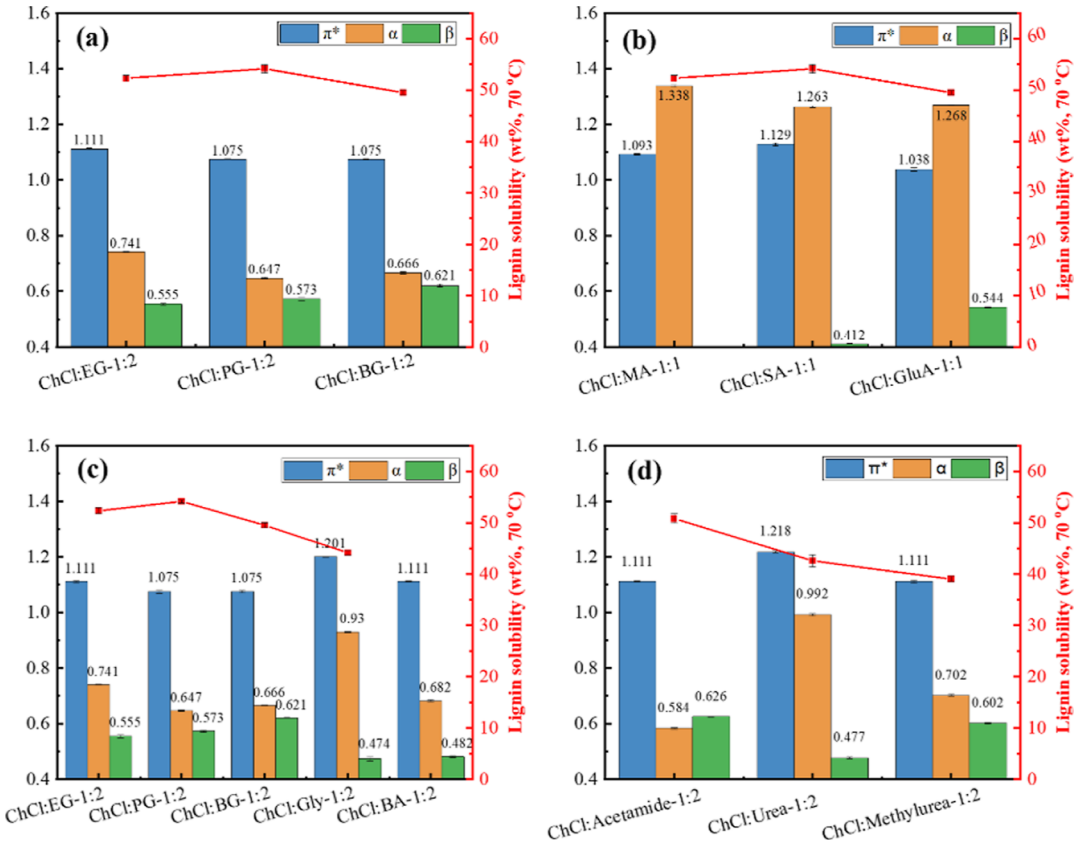


Figure 4. Effect of the alkyl chain length of HBDs (a,b), and the type of ChCl-based DESs (c,d) on the values of α, β, and π^*.

urea—1:2) when the HBD was polyalcohol (e.g., EG) or amide (e.g., urea), suggesting that metal halide-based DESs had stronger polarity. Especially, $ZnCl_2$/urea—1:4 showed the largest value of π^* (1.472). In another aspect, when the HBD was EG or urea, $ZnCl_2$-based DESs showed higher hydrogen bond acidity than ChCl-based DESs, while the hydrogen bond acidity of $ZnCl_2$/LA—1:2 was lower than that of ChCl/LA—1:2. More importantly, for the investigated ammonium salt and metal halide-based DESs, the change tendency for hydrogen bond basicity (β) was opposite with those for α and π^*. The property of having larger polarity or acidity made metal halide-based DESs a promising reference to explore new DESs with such requirements.

Subsequently, it was observed that the alkyl chain length in the HBAs could affect the α, β and π^* values. Three DESs, i.e., TMAC/EG—1:4, TEAC/EG—1:4, and TBAC/EG—1:4, were selected to analyze the effect of alkyl chain length of HBAs on α, β, and π^* values. With the increase of the alkyl chain length, the value of π^* showed a downward trend (Figure 3a): TMAC/EG—1:4 > TEAC/EG—1:4 > TBAC/EG—1:4, implying that less non-specific interactions occurred between the DESs and the dye with an increased alkyl chain length because π^* mainly reflected non-specific interactions between DES and dye.[27] In other words, longer alkyl chain of the quaternary ammonium salts in DESs would result in lower polarity or polarization. In contrast, an opposite trend was observed for the change of β values in the three DESs because the cation−anion interaction in HBAs weakened when increasing the alkyl chain length, making the salt anion more receptive to the protons of the HBD species.[28] However, the α

https://doi.org/10.1021/acs.iecr.3c01309
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values of the three DESs were irregular with the alkyl chain length.

Besides, the influence of the anion type in HBAs on α, β, and π^* was investigated. Four DESs, i.e., TMAC/EG—1:4, TMAA/EG—1:4, TBAC/EG—1:4, and TBAB/EG—1:4, were employed to discuss the relationship between the anion types of HBAs and the Kamlet−Taft parameters. As shown in Figure 3b, the α values of the four DESs were irregular and slightly different. Especially, the α value of TMAA/EG—1:4 was relatively low due to the higher basicity of acetate (Ac^-) than Cl^-. These results indicated that TMAA/EG—1:4 had the weakest hydrogen bond-donating ability among the four DESs. Moreover, the β value was mainly affected by the nature of the anion in the HBAs. TMAC/EG—1:4 has a larger β value than TMAA/EG—1:4. Generally, Cl^- has a stronger ability to accept hydrogen atom than acetate (Ac^-), resulting from the fact that Cl^- has a stronger ability to form hydrogen bonds than Ac^-. Thus, TMAC/EG—1:4 had a larger value of β. Meanwhile, for TBAC/EG—1:4 and TBAB/EG—1:4, the values of β were not much different, indicating that Cl^- and Br^- have similar proton-accepting ability. Additionally, the change on the value of π^* followed a similar tendency with that for β value.

3.2.2. Effect of HBDs. Herein, 18 kinds of ChCl-based DESs were taken as the representative to discuss and analyze the effect of HBDs on the values of α, β, and π^* (Figure 4). Similar to HBAs, the length of the alkyl chain in the HBDs also has strong impact on α, β, and π^*. Three ChCl−polyalcohol [i.e., EG, 1,2-propylene glycol (PG), and 1,2-butanediol (BG)] DESs and three ChCl−acid [i.e., malonic acid (MA), succinic acid (SA), and glutaric acid (GluA)] DESs were employed to discuss the influence of alkyl chain length of HBDs. As shown in Figure 4a,b, the value of β was boosted by increasing the alkyl chain length of HBDs for both ChCl−polyalcohol or ChCl−acid DESs, resulting from the weakened cation−anion interaction with the increase of the alkyl chain, which was consistent with the previous reports that the hydrogen bond basicity of DESs depended on the alkyl chain length of the acid.[29,30] Moreover, the α value of ChCl−acid DESs were much larger than that of ChCl−polyalcohol DESs (Figure 4a,b), originating from the acidic nature of ChCl−acid DESs.[27] Besides, with the increase of the chain length of the organic acids, the α value of the DESs tended to decrease slightly, resulting from the fact that the acidity of the organic acids decreased with the increased alkyl chain length. /

Furthermore, the α, β, and π^* of ChCl-based DESs (i.e., ChCl−polyalcohol DESs, ChCl−amide DESs, and ChCl−acid DESs) were discussed separately. First, we employed five alcohols as HBDs, including EG, PG, BG, glycerol (Gly), and benzyl alcohol (BA), at a ChCl/HBD molar ratio of 1:2. Obviously, the α value of the synthesized ChCl−polyalcohol DESs increased with the increasing number of hydroxyl groups in alcohols (Figure 4c) because more hydroxyl groups provided more sites to donate hydrogen bonds, and ChCl/Gly—1:2 has the largest α value (0.93). Meanwhile, ChCl/Gly—1:2 possesses the highest π^* value (1.201) because more hydroxyl groups resulted in its high polarity and the strong ability to dipole−dipole and dipole-induced dipole interactions. In contrast, the β value (0.474) of ChCl/Gly—1:2 is the smallest, indicating its weakest ability to accept hydrogen. Additionally, the values of α, β, and π^* for ChCl/BA—1:2 show a slight difference with other polyalcohol-based DESs owing to the electronic effect of benzene ring. Second, three ChCl−amide DESs were prepared to employ urea, methyl urea, and acetamide as HBDs at a ChCl/HBD molar ratio of 1:2. Among these DESs (Figure 4d), ChCl/urea—1:2 has the highest α (0.992) and π^* (1.218) values but the lowest β (0.447) value. Moreover, urea and methyl urea contained more −NH groups, resulting in the stronger hydrogen bond donating ability (higher α values) of ChCl/urea—1:2 and ChCl/methyl urea—1:2 than ChCl/acetamide—1:2. For ChCl/urea—1:2 and ChCl/methyl urea—1:2, the methyl group resulted in a decrease in the ability to donate hydrogens; thus, ChCl/methyl urea—1:2 has a lower α value (0.702). Besides, the methyl group also made ChCl/methyl urea—1:2 have a lower π^* (1.111) value than ChCl/urea—1:2. Additionally, ChCl/urea—1:2 has the lowest β value (0.477). Third, 10 kinds of ChCl−acid DESs were prepared using 10 carboxylic acids, including formic acid (FA), acetic acid (AA), LA, MA, SA, GluA, levulinic acid (LeA), acrylic acid (AcA), glycolic acid (GlyA), and OA. In these DESs, the molar ratio of ChCl and carboxylic acids is 1:2 except for the DES by using SA as the HBD. The molar ratio of ChCl and SA was 1:1 because a DES could not be formed with a ChCl/SA molar ratio of 1:2. As shown in Figure 5, the π^* values of ChCl−acid DESs are similar (from 0.992 to 1.138). In contrast, the α values of these DESs ranged from 0.914 to 1.333 except for ChCl/OA—1:2, which has the highest α value of 2.053. Meanwhile, these DESs had β values in the range of 0.171−0.495, except that ChCl/OA (1:2) has the larger β (1.262) value.

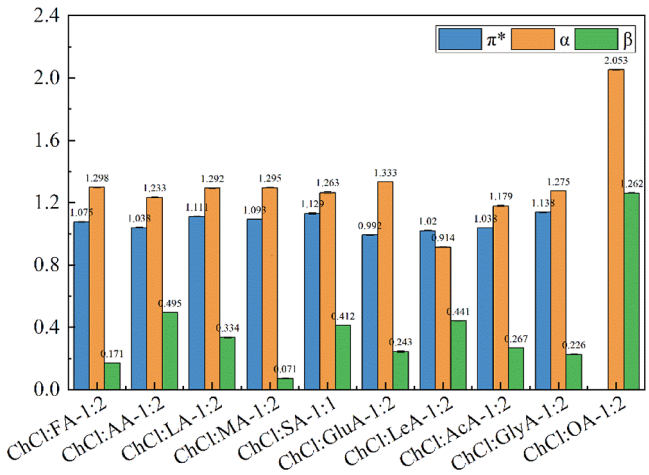


Figure 5. α, β, and π^* values of ChCl−acid DESs.

Besides the effect of HBAs and HBDs, it should be pointed out that measurement conditions generally have significant impact on the Kamlet−Taft parameters.[22,31,32] First, for the same DES, the values of Kamlet−Taft parameters significantly varied with the water contents. For example, the β value of ChCl/OA (1:1) is 1.825 in this work, while it is 0.825[31] because the employed solvent is a 70 wt % aqueous solution of ChCl/OA (1:1). It has been reported that too high content of water would significantly reduce the β value.[22,33] Second, the types of the used dyes could also affect the determined values of the Kamlet−Taft solvatochromic parameters because of the different intermolecular interactions between the dye and the DES. To ensure the comparability, the same dyes were employed when comparing the Kamlet−Taft parameters in this work with the literature studies.[34] Therefore, the

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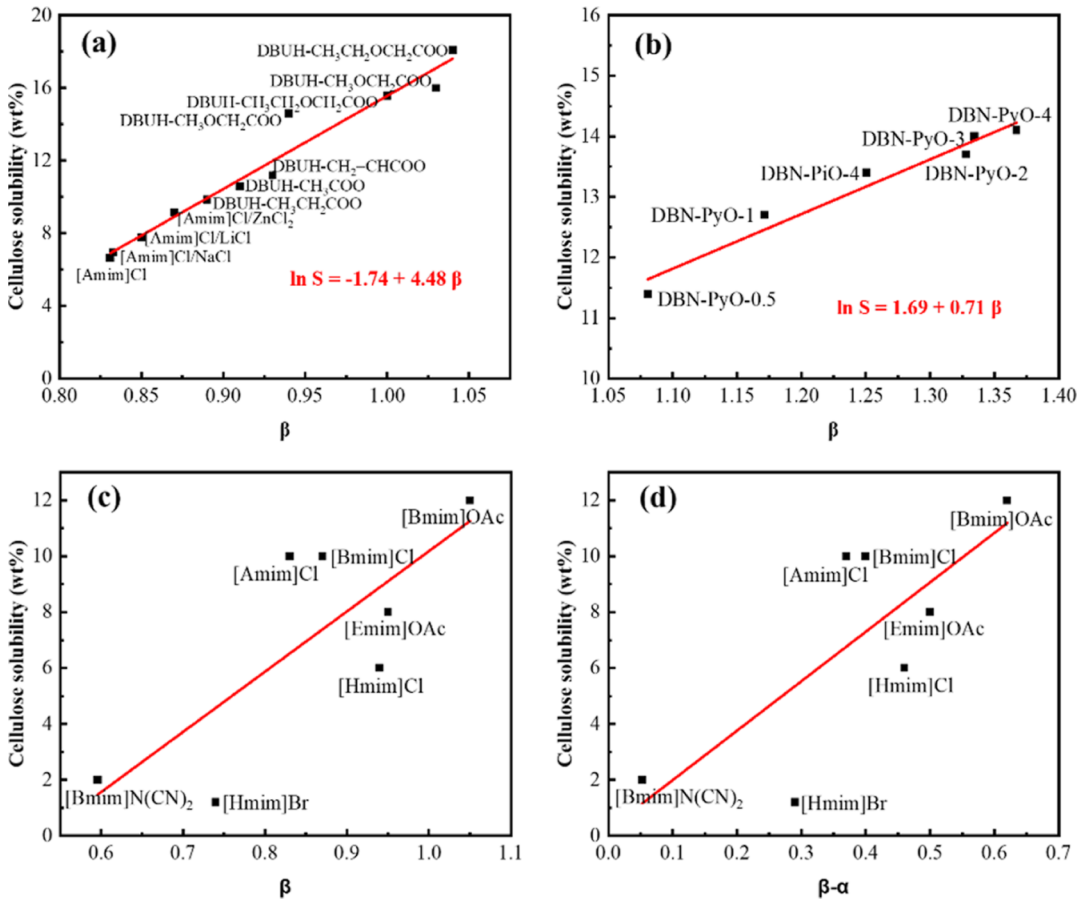


Figure 6. Relationship between cellulose solubility and Kamlet−Taft solvatochromic parameters in the literature studies: (a) 80 °C, (b) 70 °C, (c) 100 °C, and (d) 100 °C.

measurement conditions were controlled carefully to ensure the comparability of the achieved values of Kamlet−Taft parameters. Additionally, the determination temperature generally has no impact on β and π^* values,[33] and the change tendency is similar for the α value of different DESs at different temperature. In this work, the Kamlet−Taft parameters of the prepared DESs were discussed in categories to reduce the influence of their determined conditions. Considering the potential impact of the determined conditions on the Kamlet−Taft parameters, it was worth paying more attention on this aspect in future works.

3.3. Comparison Kamlet−Taft Solvatochromic Parameters of DESs, ILs, and Molecular Solvents. For comparison, the Kamlet−Taft parameters of some common molecular solvents and ILs were collected from a typical literature[35] and are also provided in Tables S3 and S4. Considering that the Kamlet−Taft parameters of the prepared DESs were determined based on the reported method, the obtained results for the DESs were comparable with the data of common molecular solvents and ILs from the literature studies. It was found that water had larger α and π^* values but a relatively smaller β value than most of other molecular solvents. However, there were great differences on the values of Kamlet−Taft parameters for water in different literature studies. More importantly, the synthesized DESs have relatively higher Kamlet−Taft parameters (α, β, and π^*) than molecular solvents (Tables S2 and S3), which is consistent with the result that the α value of DESs is higher than that of most organic solvents predicted using the COSMO-RS model.[36] Additionally, the Kamlet−Taft parameters of ILs are larger than those of molecular solvents (Tables S3 and S4), and protic ILs have larger α and β values than aprotic ILs, suggesting that protic ILs have stronger hydrogen bond-donating and hydrogen bond-accepting abilities.

From the results in Tables S2−S4, we could conclude that most of the developed DESs have relatively larger Kamlet−Taft parameters (α, β, and π^*) than common molecular solvents, while DESs have a larger α value, slightly smaller β and ($\beta-\alpha$) values, and comparable π^* value in comparison with ILs. Furthermore, we also attempted to compare some of our results with reported values. Unfortunately, it was not easy to find more systems for comparison since the reported Kamlet−Taft parameters of DESs were determined at different conditions, e.g., the molar ratio, the type of dye probes, and the calculation formula. For example, Florindo et al. selected three probes, i.e., Reichard'33, NEt_2, and NH_2, to measure the Kamlet−Taft parameters of DESs,[22] which is different from the NR used in our work. Thereby, there is a small difference on the β and π^* values in the literature studies and in our work, while the difference on α values is a bit large because the interactions between different dye probes and DESs were different, thus causing the inconsistence of data from different sources. Moreover, several 70 wt % DES aqueous solutions had relatively larger α, β, and π^* values than our determined results.[31] Therefore, when comparing the Kamlet−Taft parameters of different solvents, the determined conditions should be paid more attention.

https://doi.org/10.1021/acs.iecr.3c01309
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Table 1. Solubility of Cellulose in Different DESs and ILs

entry	solvents	β	$\beta-\alpha$	condition (°C)	cellulose type	solubility (wt %)	ref
1	TMAC/FA—1:3	1.094	−0.876	90	MCC	<0.49	this work
2	TMAC/GlyA—1:2	1.288	−0.715	90	MCC	<0.54	this work
3	TBAC/EG—1:2	0.805	0.096	90	MCC	<0.66	this work
4	ChCl/OA—1:2	1.262	−0.791	90	MCC	<0.58	this work
5	ChCl/OA—1:1	1.136	−0.928	90	MCC	<0.62	this work
6	[Amim]Cl	0.83	0.37	100	MCC	~10	45
7	[Bmim]Cl	0.87	0.40	100	pulp cellulose	10	46, 47
8	[Hmim]Cl	0.94	0.46	100	Avicel	6	48, 49
9	[Hmim]Br	0.74	0.29	100	Avicel	1−2	48, 49
10	[Bmim]OAc	1.05	0.62	100	Avicel	12	49
11	[Emim][OAc]	0.95	0.55	100	Avicel	8	49, 50
12	[Bmim]$N(CN)_2$	0.596	0.053	100	dry cellulose	<2	51
13	[Bmim]NTf_2	0.243	−0.374	110	Avicel	<0.553	52
14	AmimCl	0.831		80	cotton cellulose	6.64	39
15	AmimCl/NaCl—1 wt %	0.833		80	cotton cellulose	6.93	39
16	AmimCl/LiCl—2 wt %	0.85		80	cotton cellulose	7.76	39
17	AmimCl/$ZnCl_2$—1 wt %	0.87		80	cotton cellulose	9.14	39
18	[DBUH][CH_3CH_2COO]	0.89	0.40	80	MCC	9.85	37
19	[DBUH][CH_3COO]	0.91	0.41	80	MCC	10.57	37
20	[DBUH][CH_2═$CHCOO$]	0.93	0.45	80	MCC	11.18	37
21	[DBNH][CH_3OCH_2COO]	0.94	0.53	80	MCC	14.6	37
22	[DBNH][$CH_3CH_2OCH_2COO$]	1.00	0.58	80	MCC	15.55	25
23	[DBUH][CH_3OCH_2COO]	1.03	0.52	80	MCC	15.98	25
24	[DBUH][$CH_3CH_2OCH_2COO$]	1.04	0.51	80	MCC	18.08	25

Besides, some Kamlet−Taft parameters of DESs that were not mentioned above could be found in Figure S2 and S3. In Figure S2, the α value of betaine−polyalcohol DESs is slightly lower than that of betaine−acid DESs, but the β value of betaine−polyalcohol DESs is relatively higher. For betaine−polyalcohol DESs, the α and π^* values of betaine/Gly—1:2 are the highest, while the β value is the lowest, which may be caused by the number of hydroxyls in Gly. In addition, the α value of betaine/FA (1:3) and TMAC/FA (1:3) is higher. These phenomena are consistent with the results of ChCl−polyalcohol DESs and ChCl−acid DESs. In Figure S3, it can be observed that for $MgCl_2{\cdot}6H_2O$/alcohol/urea DESs, the β value is low, but the π^* value is higher, which is consistent with the conclusion that the water content in the solvent systems increased the polarity/polarization but decreased the hydrogen bond-accepting ability.

3.4. Kamlet−Taft Solvatochromic Parameters of DESs and Solubility of Lignocellulosic Components.

3.4.1. Kamlet−Taft Solvatochromic Parameters and Cellulose Solubility. Generally, the solubility of cellulose has a very good positive linear correlation with the β value of the solvent, and the resources and the molecular weight of the cellulose showed negligible impact on this correlation.[9,37] As reported, to dissolve the cellulose well, the β values of ILs should range from 0.80 to 1.20, whereas the net basicity, which could be reflected by the value of $\beta-\alpha$, should neither be lower than 0.35 nor exceed 0.90, reflecting a necessary balance between acidity and basicity.[38] Herein, the solubility of cellulose in ILs was fitted the LSER through multiple linear regression to determine the correlation between cellulose solubility and Kamlet−Taft parameters,[37,39] and it could be observed that the solubility of cellulose has a positive linear relationship with the β value (Figure 6a,b). Besides, several ILs could well dissolve the cellulose (Table 1, entries 6−8, 10, and 11); the β values of these ILs are mainly around 0.9, and their $\beta-\alpha$ values are around 0.4. However, there is no linear relationship between the Kamlet−Taft parameters and cellulose solubility in these ILs (Figure 6c,d), presumably due to different types of cellulose. Additionally, [Bmim]$N(CN)_2$ and [Hmim]Br have lower β values (0.596 and 0.74, respectively), and their α values are only 0.053 and 0.29. Thus, the solubility of cellulose in these two ILs is very poor (Figure 7).

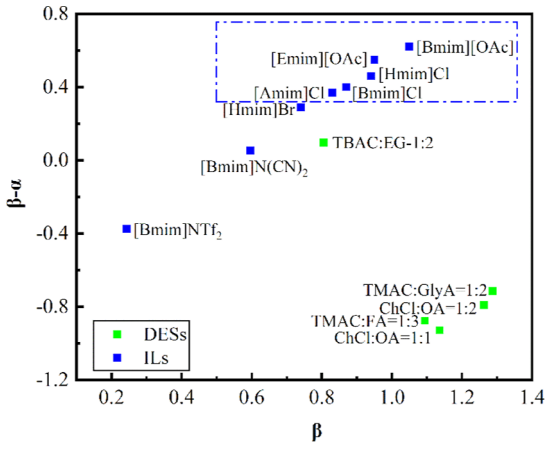


Figure 7. β and $\beta-\alpha$ values of the DESs and ILs.

The dissolution of cellulose in ILs has been widely reported, and the relationship between cellulose solubility and the IL's Kamlet−Taft parameters has also been well investigated. However, very little effort was devoted on revealing the relationship between the solubility of cellulose in DESs and the Kamlet−Taft parameters. Ren et al. synthesized and characterized a series of ChCl-based DESs and investigated their cellulose solubility.[40] The results showed that the β value has a significant correlation with the solubility in ChCl−DESs.

https://doi.org/10.1021/acs.iecr.3c01309
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Therefore, we speculated that the solubility of cellulose in DESs may be related to the β value. Compared with ILs,[41] the solubility of cellulose in traditional DESs was lower. There was one hydroxyl (−OH) group in the choline cation, which would compete with cellulose to form a hydrogen bond with chlorine, resulting in poor solubility of cellulose (usually less than 1.0 wt %). Thus, to improve the solubility of cellulose in DESs, the hydrogen bond-accepting ability of DESs should be improved first.[42,43] In order to clarify the relationship between the Kamlet−Taft parameters (especially β) of DESs and cellulose solubility, several DESs with higher β value (>0.8) in this work were selected to measure the cellulose solubility (Table 1), and the results are shown in Table 1 and Figure 7. Although the β values of these DESs are high (around 1.2), they hardly dissolved cellulose owing to their too low $\beta-\alpha$ values (less than 0). Therefore, we speculated that if DESs could efficiently dissolve cellulose, they should have both suitable β and $\beta-\alpha$ values, which was consistent with the phenomenon observed with ILs as the solvents. For the dissolution of cellulose in ILs, the β and $\beta-\alpha$ values should be located within a certain section: $0.8 < \beta < 1.2$ and $0.35 < \beta-\alpha < 0.95$, respectively. Owing to the greater β values (>0.8) of the investigated DESs, the DES should have relatively lower ability to donate hydrogen bonding in order to better dissolve cellulose. In other words, the DES with stronger basicity could potentially dissolve cellulose efficiently, which agreed with our previous results.[9]

Considering the low solubility of cellulose in the used DESs, the changes in cellulose after being dissolved and regenerated were not investigated in this work. However, Clough et al. have pointed out that it is not always an innocent dissolution process or reaction that could be expected, at least in some cases in ILs,[44] and so is the case of DESs. In the future, the influence of DESs on cellulose properties (i.e., molecular weight and crystalline allomorph) should be necessarily investigated for the designed new DESs of dissolving cellulose. Meanwhile, some types of DESs may lead to the formation of side products (e.g., 5-hydroxymethylfurfural), which also required more attention. More importantly, more efforts should be devoted on designing innovative DESs with good ability to dissolve cellulose and exploring the correlation between cellulose solubility and their Kamlet−Taft parameters, which would be of practical significance.

3.4.2. Kamlet−Taft Solvatochromic Parameters of DESs and Solubility of Kraft Lignin. We analyzed the relationship between the Kamlet−Taft solvatochromic parameters of our previously reported solvents and the solubility of lignin, cellulose, and hemicellulose.[9,11,20] Considering that the resources and structures of lignin significantly affected the correlation between lignin solubility and the Kamlet−Taft parameters, herein, we employed Kraft lignin as the representative to investigate the correlation. Based on the results in Table S7 and Figure 6b, it was found that the solvent system for dissolving Kraft lignin had a positive linear relationship with α or π^* values, but not with the β value.

In this work, the solubilities of Kraft lignin in ChCl−polyalcohol, ChCl−acid, ChCl−amide, and quaternary ammonium salt−EG DESs were measured (Table 2 and Figures 3 and 4). As shown in Figure 8, a positive linear correlation was observed between the solubility of Kraft lignin and the $\alpha-\beta$ values of some of the developed DESs, while an inverse correlation between the solubility of Kraft lignin and their β values. However, the linear correlation between Kraft lignin solubility and Kamlet−Taft parameters was not as good as the correlation between cellulose solubility and β values, which was a common phenomenon for lignin dissolution.[53] Besides, the lignin type significantly affected the solubility and the correlation between the solubility and Kamlet−Taft solvatochromic parameters. Herein, we only discuss the influence of DES types on the solubility of Kraft lignin, and more jobs are undergoing to determine the solubility of various types of lignin and analyze the correlation between solubility and DES types.

Table 2. Solubility of Lignin and Xylan in Several DESs

entry	DESs	molar ratio	solubility (wt %, 70 °C)[a,b]
		Lignin	
1	TMAC/EG	1:4	54.65 (±0.63)
2	TEAC/EG	1:4	54.19 (±0.36)
3	TBAC/EG	1:4	53.84 (±0.85)
4	TBAB/EG	1:4	17.32 (±1.03)
5	TMAA/EG	1:4	55.96 (±0.92)
6	ChCl/EG	1:2	52.32 (±0.61)
7	ChCl/PG	1:2	54.15 (±0.33)
8	ChCl/BG	1:2	49.51 (±0.55)
9	ChCl/Gly	1:2	44.13 (±0.48)
10	ChCl/MA	1:1	41.99 (±0.61)
11	ChCl/SA	1:1	26.30 (±0.72)
12	ChCl/GluA	1:1	4.08 (±0.21)
13	ChCl/acetamide	1:2	50.81 (±0.91)
14	ChCl/urea	1:2	42.57 (±1.22)
15	ChCl/methyl urea	1:2	39.01 (±0.50)
		Xylan	
16	ChCl/EG	1:2	15.81 (±0.30)
17	ChCl/Gly	1:2	14.14 (±0.57)
18	ChCl/LA	1:2	19.8 (±0.24)
19	ChCl/urea	1:2	23.73 (±0.66)
20	Bet/EG	1:4	17.44 (±0.15)
21	TBAC/EG	1:2	24.83 (±0.36)
22	ChCl/GlyA	1:2	15.47 (±0.45)

[a]The standard uncertainty for the temperature was $u(T) = 0.2$ °C. [b]The values of solubility were the average of twice measurements, and the values in parentheses represented half of the range of replicate results. The uncertainties for the solubilities mainly resulted from the measurement process and sample weighting.

Among the 56 kinds of DESs investigated in this work, ChCl/OA—1:2 had the most significant value of β (1.262), and the β value of TMAC/FA—1:3 could reach up to 1.149 (Figure S3). It was speculated that these two DES systems might have high efficiency for biomass pretreatment. There were many reports on biomass pretreatment with the ChCl/OA DES system. For example, Husanu et al. evaluated the ability of a series of DESs for extraction of high value-added compounds from chestnut shell waste[54] and found that the extraction efficiency of acid-based DESs, sugar-based DESs, and polyol-based DESs decreased successively, which was well correlated with the trend of Kamlet−Taft parameter of α value. Kohli et al. used microwave-assisted DESs to remove lignin from lignocellulose effectively. They found that the ChCl/OA and ChCl/FA systems had the highest removal efficiency on the birch wood and miscanthus lignin content.[51] For some reported DESs, the solubility of lignin was related to α and π^* values, and it had a good linear correlation. However, as discussed above, we found a good linear correlation between lignin solubility and the $\alpha-\beta$ values for ChCl−acid DESs. In

https://doi.org/10.1021/acs.iecr.3c01309
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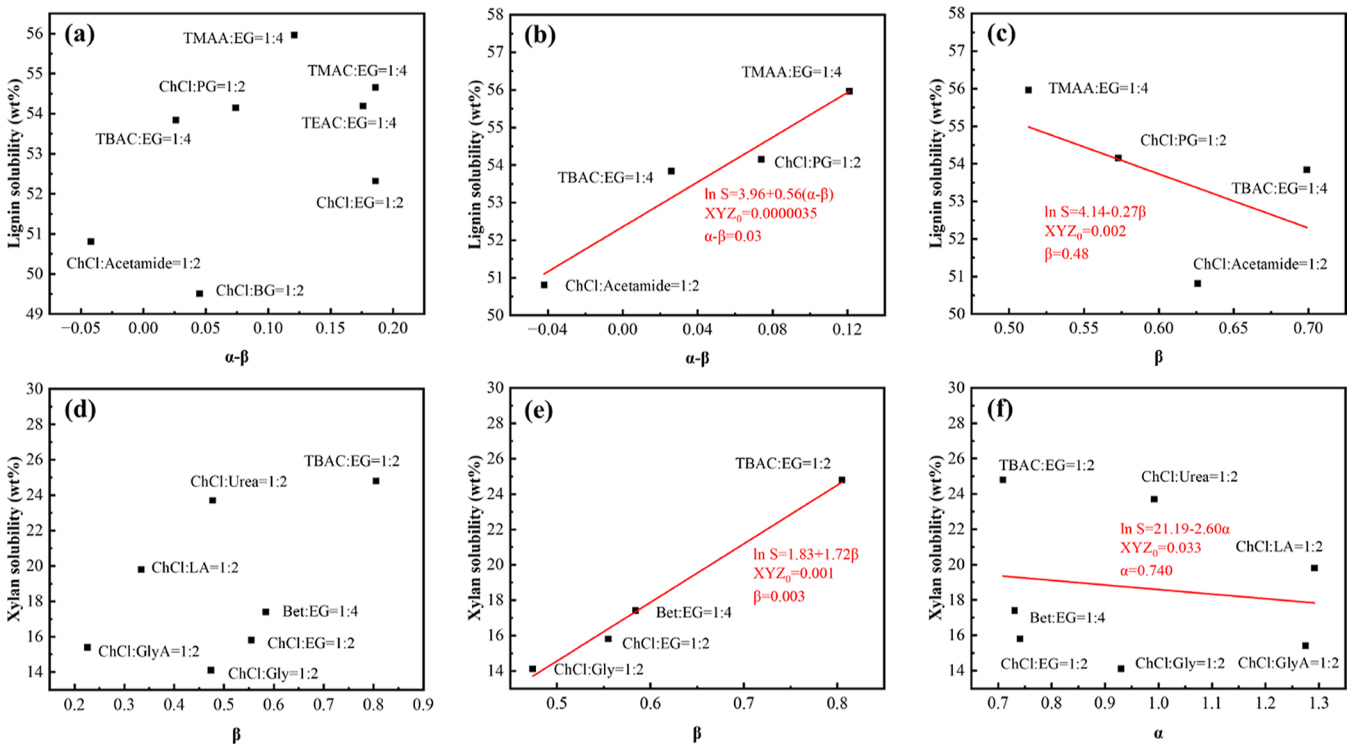


Figure 8. Relationship between the Kamlet−Taft parameters of the examined DESs and the corresponding solubilities of lignin (a−c) or xylan (d−f).

contrast, an inverse correlation was observed between lignin solubility and the β values, although the linear correlation was not noticeable. These results indicated that the ability of the DES to donate hydrogen bonding showed higher impact on the lignin dissolution than the ability to accept hydrogen bonding. That is to say, to dissolve lignin efficiently, the DES should have a more vital ability to donate hydrogen bonding to lignin. Unfortunately, the linear relationship between lignin solubility and Kamlet−Taft parameters was poor in other types of the developed 56 DESs. More effort was required to reveal the true relationship between the Kamlet−Taft parameters of DESs and lignin solubility. Based on these preliminary results, several novel DESs have been designed to dissolve lignin (the results were still optimized in our lab), confirming the applicability of the preliminary regularity for designing efficient DESs. Despite this finding, not all DESs followed the same regularity. When constructing suitable DESs to dissolve lignin, the types of DESs and lignin should also be considered.

Besides, the regeneration of Kraft lignin from the four DESs, i.e., ChCl/EG—1:2, TMAC/EG—1:2, ChCl/MA—1:1, and ChCl/acetamide—1:2, was investigated using ethanol as the anti-solvent, and the recovered yields of the dissolved lignin from these four DESs were all above 94%. The regenerated lignin was characterized by 2D HSQC NMR (Figures S4 and S5), which was compared with the result for the original lignin. Obviously, most of the lignin structures were relatively intact. Importantly, for the regenerated lignin from all the examined DESs, only the β-O-4 linkages were partially destroyed during the process of dissolution and regeneration, thereby decreasing the molecular weight of the investigated lignin. It is highly important to investigate the effect of DESs on the properties of the regenerated lignin, which affects the specific application of lignin. Thereby, this aspect should be paid particular attention when selecting DESs to dissolve lignin.

3.4.3. Relationship between the Kamlet−Taft Solvatochromic Parameters of DESs and Xylan Solubility. Previous studies have shown that it was an effective strategy for screening DESs to dissociate hemicellulose from lignocellulose based on the solubility of xylan.[55,56] Morais et al. found that an aqueous solution of ChCl/urea—1:2 with a concentration of 66.7 wt % showed high ability to dissolve xylan, and the solubility could reach 328.23 g/L at 80 °C.[57] Colombo Dugoni et al. prepared DESs with choline acetate (ChOAc) as the HBA and imidazole (Im) as the HBD for hemicellulose dissolution. They observed that the solubility of hemicellulose could reach 35 wt % in the developed DESs without water,[58] and the maximum solubility could reach up to 45 wt % in an aqueous solution of ChOAc/Im with a concentration of 85 wt % at 80 °C. These studies confirmed that DESs were potential solvents for hemicellulose dissolution. Moreover, Yu et al. reported that the solubility of xylan was positively correlated with the β value of DESs.[11] In contrast, Yang et al. used alkaline DESs to dissolve xylan. However, the xylan solubility was not significantly correlated with the values of α, β, π^*, and even their combinations ($p > 0.05$).[59] The different types of the employed DESs might cause the difference. Herein, the solubilities of xylan in several DESs were measured (Table 2, entries 16−22). Obviously, TBAC/EG—1:2 showed the best performance on dissolving xylan, and the solubility could reach 24.8 wt % at 70 °C. In comparison, ChCl/Gly—1:2 provided the lowest solubility (only 14.1 wt %). In addition, multivariate linear fitting was carried out to analyze the relationship between the solubility of xylan and the Kamlet−Taft solvatochromic parameters (Figure 8). Overall, the DESs with a larger β value had a higher xylan solubility, representing that the DES should be able to accept hydrogen bonding from hemicellulose to dissolve hemicellulose efficiently. Especially, there was a good positive linear correlation between the β

https://doi.org/10.1021/acs.iecr.3c01309
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value of alcohol-based DESs and the xylan solubility. Despite this finding, it did not necessarily mean that all DESs obeyed the same rule. When selecting suitable DESs to dissolve hemicellulose, the types of DESs and the properties of hemicellulose should also be considered.

3.5. Effect of Different Interactions on the Correlation between Kamlet−Taft Parameters and Solubility. As discussed above, the dissolution ability of one DES was different for lignin, cellulose, and hemicellulose, which was probably caused by the different mechanisms of the interaction between DES and different lignocellulosic components. First, lignin has plenty of intramolecular hydrogen bonding interactions and $\pi\cdots\pi$ stacking interactions. Breaking these two interactions is necessary to dissolve lignin in a solvent. Generally, two main non-covalent interactions, i.e., C−H$\cdots\pi$ and O$\cdots$H bonding, could be formed between DESs and lignin. The formed interactions between DESs and lignin could efficiently destroy the intramolecular interactions of lignin owing to their greater interaction energy than that between lignin molecules, resulting in the successful dissolution of lignin in DESs. Although no $\pi\cdots\pi$ stacking interaction was generated between DESs and lignin, the strong O$\cdots$H bonding can promote the dissolution of lignin in DESs.[60] Second, the difficulty to dissolve cellulose and hemicellulose in conventional solvents originated from the intramolecular and intermolecular hydrogen bonding interactions. Both HBDs and HBAs in DESs have sites to form hydrogen bonding interactions with the chains of cellulose and hemicellulose. Thus, during the dissolving of cellulose and hemicellulose in DESs, HBAs could form new hydrogen bonding with the −OH group and $-CH_2OH$ on cellulose and hemicellulose chains, while HBDs would form new hydrogen bonding with skeletal oxygen and −OH group on cellulose and hemicellulose chains. The synergistic effect of HBDs and HBAs leads to the destruction of the hydrogen bond network structure of cellulose and hemicellulose, thus resulting in the dissolution of cellulose and hemicellulose in DESs.[61] From these discussions above, it could be observed that the interactions between DESs and lignin were complex, thus making the correlation between Kamlet−Taft parameters and the lignin solubility be unsatisfactory. In comparison, the interactions between DESs and cellulose/hemicellulose was only the hydrogen bonding interactions. Therefore, the solubility of cellulose/hemicellulose showed a good correlation with the β value for most of the developed DESs. It would be an interesting aspect to evaluate the correlation between Kamlet−Taft parameters and the solubility of the lignocellulosic components.

4. CONCLUSIONS

Owing to their unique properties, DESs are up-and-coming alternatives to organic solvents, showing apparent advantages compared to ILs. The Kamlet−Taft solvatochromic parameters can be used to characterize the properties of DESs. By analyzing the relationship between the Kamlet−Taft solvatochromic parameters of 56 DESs, the following conclusions were drawn:

(1) The Kamlet−Taft solvatochromic parameters of DESs were mainly related to the structures and properties of HBAs and HBDs. For example, the increase in the length of the alkyl chain would reduce the ability of the target substance to provide or accept hydrogen; thus, the values of both α and β decreased with the increase in the alkyl chain length. The π^* parameter was mainly related to the non-specific interaction between DESs and dyes. In addition, the water content significantly affected the Kamlet−Taft solvatochromic parameters of DESs.

(2) Generally, DESs had relatively larger α and π^* values but smaller β values than common molecular solvents. In comparison with ILs, DESs had larger α values, slightly smaller β and $\beta-\alpha$ (which could represent the net basicity of the solvents) values, and comparable π^* values. As reported, ILs that could efficiently dissolve cellulose should have a β value ranging from 0.80 to 1.20, and the $\beta-\alpha$ (net basicity) value should not be less than 0.35 or more than 0.90. However, the developed DESs in this work also obey this rule to some extent. Although the investigated DESs had a β value of greater than 0.8, too weak net basicity ($\beta-\alpha < 0.35$) resulted in the solubility of cellulose being very poor in these DESs. Notably, the solubility of lignin had a good linear correlation with the $\alpha-\beta$ (net acidity) values of the examined DESs, while the solubility of xylan well positively correlated with the β values of the determined DESs. However, the linear relationship between the solubility of lignin and xylan and other parameters was poor. In addition, although Kamlet−Taft solvatochromic parameters can generally be a good indication for selecting DESs to dissolve cellulose, lignin, or hemicellulose efficiently, it sometimes did not necessarily mean that the DESs with suitable Kamlet−Taft solvatochromic parameters certainly dissolved the three components of lignocellulose. Besides Kamlet−Taft solvatochromic parameters, other factors, i.e., the properties of the components and other physicochemical properties, should also be considered when choosing the suitable DESs for dissolving lignocellulosic components. Therefore, caution should be kept on this point.

Therefore, by understanding the Kamlet−Taft parameters of DESs, DESs could be reasonably designed as a suitable solvent for dissolving lignocellulose and its main components (i.e., cellulose, lignin, and hemicellulose). To fulfill this purpose, we are designing more types of DESs with diverse structures and properties, including those reported by other researchers, and are carefully determining their Kamlet−Taft parameters. By combining necessary theoretical calculations, we can obtain how the structures and properties of DESs intrinsically affect their Kamlet−Taft parameters. These developed DESs would be further applied to dissolve cellulose, hemicellulose, and lignin. It is highly promising to reveal an accurate relationship between the solubility and Kamlet−Taft parameters of DESs, which would guide the design of effective DESs to pretreat, dissolve, and even convert lignocellulose.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.iecr.3c01309.

Chemicals; procedure to prepare DESs; Kamlet−Taft solvatochromic parameter measurement; solubility measurement; Kamlet−Taft solvatochromic parameters of the developed DESs, ILs, and common molecular solvents reported in the literature studies; comparison of the Kamlet−Taft parameters of the same DES measured

https://doi.org/10.1021/acs.iecr.3c01309
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in this work and the literature studies; Kamlet−Taft parameters and LSER of solvent systems with good solubility; LSER equation of the Kamlet−Taft parameters for DESs with lignin solubility; and chemical structures of HBAs, HBDs, and three probes (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors thank the National Natural Science Foundation of China (21873012) and the Forestry and Grassland Science and Technology Innovation Youth Top Talent Project of China (no. 2020132609) for financial support.

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https://doi.org/10.1021/acs.iecr.3c01309
Ind. Eng. Chem. Res. 2023, 62, 11723−11734